

Technical Progress Report

Project name:	Biped robot		
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Start date:	2026-04-27	End date:	2026-04-30
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1. Executive Summary

This report documents the technical progress of the biped robot project in the current week. The team has completed structural adjustment, hardware compartment protection improvement, and large-scale vision dataset collection and model training. The original robot structure imposed excessive power demand that exceeded the output capacity of the current medium-size motors, so the structure was revised and leg length was shortened to reduce load and torque requirements. Meanwhile, the vision dataset was expanded to 1800 images with richer angles and scenes, significantly improving recognition accuracy and stability. During testing, the robot achieved more feasible motion performance and reliable visual identification. Typical issues including remaining structural clearance, minor gait jitter, and individual angle misidentification were recorded. The team will continue optimization to support stable, continuous, and visually guided walking in the next stage.

2. Objectives & Scope

- Complete structural redesign and leg length adjustment to match the driving capacity of medium-size motors.
- Reduce mechanical load and torque demand to ensure stable and smooth actuation of joints.
- Expand the vision dataset to a large scale and retrain the model for higher recognition success rate.
- Verify motion feasibility and vision recognition performance under the updated structure.
- Identify remaining issues in rigidity, coordination, and robustness for subsequent improvement.
- Lay a solid foundation for stable walking and integrated vision-motion control.

3. Technical Work Completed

3.1 Structural Redesign and Adjustment

The original robot structure was found to demand excessive motor power, which could not be satisfied by the existing medium-size motors. The team revised the mechanical structure and shortened the leg length to reduce

moment load and required torque, making the actuation system match the motor performance. The adjusted structure was assembled and checked to ensure dimensional consistency and connection reliability.

3.2 Hardware and Actuation Optimization

Reinforced key joints and transmission parts to improve overall rigidity. Further protected the hardware compartment to reduce exposure of the ESP32, camera, battery, and wiring, enhancing anti-drop and anti-vibration performance during movement.

3.3 Large-Scale Dataset Collection and Model Training

Collected and organized a high-volume vision dataset of 1800 images covering diverse angles, distances, lighting conditions, and positions. The dataset was used to retrain the recognition model, greatly improving feature learning ability and generalization. The model was deployed and validated in real-robot tests.

3.4 Integrated System Verification

Conducted power-on tests and basic motion trials with the updated structure. Verified motion smoothness, joint response, and real-time vision recognition under actual working conditions. Recorded motion stability and recognition success rate for quantitative evaluation.

4. Observations & Findings

- The original structure required excessively high motor output torque, which exceeded the limit of medium-size motors; shortening leg length effectively reduced load and improved motion feasibility.
- With a dataset expanded to 1800 images, the vision model achieved significantly higher recognition success rate and stronger robustness.
- Similar objects such as cubes and pyramids are still occasionally confused at extreme angles, but the frequency is greatly reduced.
- Minor jitter and lag exist in individual joints due to remaining structural clearance.
- Hardware protection is basically in place, but further enclosure can improve reliability during continuous movement.

5.Challenges & Roadblocks

- The structural adjustment improves motor adaptability, but residual clearance in joints still affects motion stability.
- Even with enhanced recognition performance, extreme viewing angles still lead to occasional classification errors.
- Coordination between left and right legs requires finer parameter tuning for continuous walking.
- The balance between compact structure, leg length, and support rigidity needs further comprehensive optimization.

6. Next Steps & Recommendations

- Further optimize joint clearance and fastening to improve motion rigidity and reduce jitter.
- Fine-tune gait parameters and actuation timing to achieve smoother and more coordinated walking.

- Supplement a small number of extreme-angle samples and perform lightweight fine-tuning on the trained model to approach near-perfect recognition.
- Fully enclose the hardware compartment for all-round protection of electronic components.
- Conduct integrated tests of stable walking + real-time vision recognition to verify overall system performance.

7. Conclusion

In the current week, the team solved the core problem of mismatched motor power by redesigning the structure and shortening leg length, making the robot motion feasible with existing medium-size motors. Meanwhile, the vision model was trained with 1800 images, which greatly improved recognition success rate and stability. The updated mechanical structure and vision system have passed initial verification, and the overall performance has been significantly improved. Remaining problems mainly include minor joint jitter, individual angle misidentification, and incomplete hardware enclosure. In the next stage, the team will focus on joint optimization, gait refinement, dataset fine-tuning, and full protection design, so as to realize stable continuous walking and high-precision visual recognition for the biped robot.